

Aggregate Expenditure and Output in the Short Run

ECON 101 – Macroeconomics

Muhebullah Karimzada

Fall 2026

Chapter Outline

- **8.1** The Aggregate Expenditure Model
- 8.2 Determining Aggregate Expenditure in the Economy
- 8.3 Graphing Macroeconomic Equilibrium
- 8.4 The Multiplier Effect
- 8.5 The Aggregate Demand Curve
- **Appendix** The Algebra of Macroeconomic Equilibrium

Chapter Outline

- **8.1** The Aggregate Expenditure Model
- **8.2** Determining Aggregate Expenditure in the Economy
- 8.3 Graphing Macroeconomic Equilibrium
- 8.4 The Multiplier Effect
- 8.5 The Aggregate Demand Curve
- **Appendix** The Algebra of Macroeconomic Equilibrium

Chapter Outline

- **8.1** The Aggregate Expenditure Model
- **8.2** Determining Aggregate Expenditure in the Economy
- **8.3** Graphing Macroeconomic Equilibrium
- 8.4 The Multiplier Effect
- 8.5 The Aggregate Demand Curve
- **Appendix** The Algebra of Macroeconomic Equilibrium

Chapter Outline

- **8.1** The Aggregate Expenditure Model
- **8.2** Determining Aggregate Expenditure in the Economy
- **8.3** Graphing Macroeconomic Equilibrium
- **8.4** The Multiplier Effect
- 8.5 The Aggregate Demand Curve
- **Appendix** The Algebra of Macroeconomic Equilibrium

Chapter Outline

- **8.1** The Aggregate Expenditure Model
- **8.2** Determining Aggregate Expenditure in the Economy
- **8.3** Graphing Macroeconomic Equilibrium
- **8.4** The Multiplier Effect
- **8.5** The Aggregate Demand Curve
- **Appendix** The Algebra of Macroeconomic Equilibrium

Chapter Outline

- **8.1** The Aggregate Expenditure Model
- **8.2** Determining Aggregate Expenditure in the Economy
- **8.3** Graphing Macroeconomic Equilibrium
- **8.4** The Multiplier Effect
- **8.5** The Aggregate Demand Curve
- **Appendix** The Algebra of Macroeconomic Equilibrium

8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build the **aggregate expenditure (AE) model**: a short-run model.
- It studies the relationship between total spending and output,
- **assuming the price level is constant** in the short run.
- Why study this? It helps us understand:
 - why recessions occur,
 - why government stimulus can boost the economy,
 - and how the **multiplier effect** works.

8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build the **aggregate expenditure (AE) model**: a short-run model.
- It studies the relationship between total spending and output,
- **assuming the price level is constant** in the short run.
- Why study this? It helps us understand:
 - why recessions occur,
 - why government stimulus can boost the economy,
 - and how the **multiplier effect** works.

8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build the **aggregate expenditure (AE) model**: a short-run model.
- It studies the relationship between total spending and output,
- **assuming the price level is constant** in the short run.
- Why study this? It helps us understand:
 - why recessions occur,
 - why government stimulus can boost the economy,
 - and how the **multiplier effect** works.

8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build the **aggregate expenditure (AE) model**: a short-run model.
- It studies the relationship between total spending and output,
- **assuming the price level is constant** in the short run.
- Why study this? It helps us understand:
 - why recessions occur,
 - why government stimulus can boost the economy,
 - and how the **multiplier effect** works.

8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build the **aggregate expenditure (AE) model**: a short-run model.
- It studies the relationship between total spending and output,
- **assuming the price level is constant** in the short run.
- Why study this? It helps us understand:
 - why recessions occur,
 - why government stimulus can boost the economy,
 - and how the **multiplier effect** works.

8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build the **aggregate expenditure (AE) model**: a short-run model.
- It studies the relationship between total spending and output,
- **assuming the price level is constant** in the short run.
- Why study this? It helps us understand:
 - why recessions occur,
 - why government stimulus can boost the economy,
 - and how the **multiplier effect** works.

8.1 Learning Objective

Goal

Understand how macroeconomic equilibrium is determined in the aggregate expenditure model.

- We build the **aggregate expenditure (AE) model**: a short-run model.
- It studies the relationship between total spending and output,
- **assuming the price level is constant** in the short run.
- Why study this? It helps us understand:
 - why recessions occur,
 - why government stimulus can boost the economy,
 - and how the **multiplier effect** works.

Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.

$$AE = C + I + G + NX$$

- C : consumption (households buying groceries, cars, Netflix subscriptions)
- I : planned investment (firms buying machines, households buying new homes)
- G : government purchases (federal, provincial, municipal)
- NX : net exports (exports minus imports)
- **Canadian AE composition (2023, Statistics Canada):**
 - $C \approx 56\%$, $I \approx 21\%$, $G \approx 22\%$, $NX \approx 1\%$ of GDP.

Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.

$$AE = C + I + G + NX$$

- **C:** consumption (households buying groceries, cars, Netflix subscriptions)
- **I:** planned investment (firms buying machines, households buying new homes)
- **G:** government purchases (federal, provincial, municipal)
- **NX:** net exports (exports minus imports)
- **Canadian AE composition (2023, Statistics Canada):**
 - $C \approx 56\%$, $I \approx 21\%$, $G \approx 22\%$, $NX \approx 1\%$ of GDP.

Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.

$$AE = C + I + G + NX$$

- C : consumption (households buying groceries, cars, Netflix subscriptions)
- I : planned investment (firms buying machines, households buying new homes)
- G : government purchases (federal, provincial, municipal)
- NX : net exports (exports minus imports)
- **Canadian AE composition (2023, Statistics Canada):**
 - $C \approx 56\%$, $I \approx 21\%$, $G \approx 22\%$, $NX \approx 1\%$ of GDP.

Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.

$$AE = C + I + G + NX$$

- C : consumption (households buying groceries, cars, Netflix subscriptions)
- I : planned investment (firms buying machines, households buying new homes)
- G : government purchases (federal, provincial, municipal)
- NX : net exports (exports minus imports)
- **Canadian AE composition (2023, Statistics Canada):**
 - $C \approx 56\%$, $I \approx 21\%$, $G \approx 22\%$, $NX \approx 1\%$ of GDP.

Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.

$$AE = C + I + G + NX$$

- C : consumption (households buying groceries, cars, Netflix subscriptions)
- I : planned investment (firms buying machines, households buying new homes)
- G : government purchases (federal, provincial, municipal)
- NX : net exports (exports minus imports)
- **Canadian AE composition (2023, Statistics Canada):**
 - $C \approx 56\%$, $I \approx 21\%$, $G \approx 22\%$, $NX \approx 1\%$ of GDP.

Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.

$$AE = C + I + G + NX$$

- C : consumption (households buying groceries, cars, Netflix subscriptions)
- I : planned investment (firms buying machines, households buying new homes)
- G : government purchases (federal, provincial, municipal)
- NX : net exports (exports minus imports)
- **Canadian AE composition (2023, Statistics Canada):**
 - $C \approx 56\%$, $I \approx 21\%$, $G \approx 22\%$, $NX \approx 1\%$ of GDP.

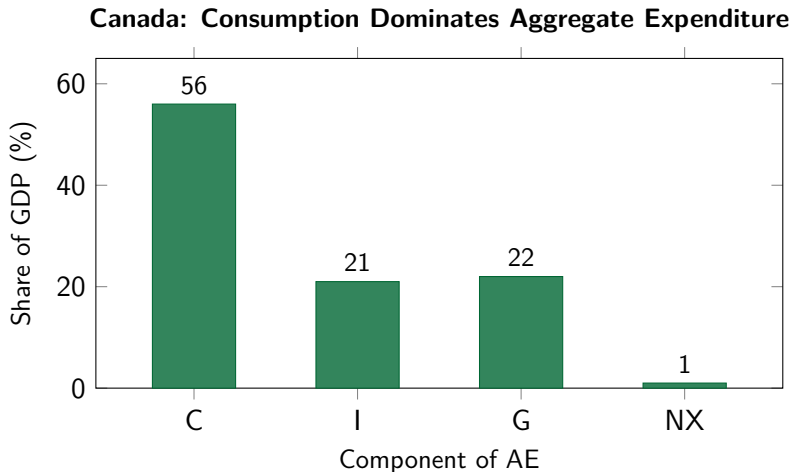
Aggregate Expenditure: Definition

- **Aggregate expenditure (AE):** total planned spending in the economy.

$$AE = C + I + G + NX$$

- C : consumption (households buying groceries, cars, Netflix subscriptions)
- I : planned investment (firms buying machines, households buying new homes)
- G : government purchases (federal, provincial, municipal)
- NX : net exports (exports minus imports)
- **Canadian AE composition (2023, Statistics Canada):**
 - $C \approx 56\%$, $I \approx 21\%$, $G \approx 22\%$, $NX \approx 1\%$ of GDP.

Table 8.1 – Canadian Components of AE (2023)



Source: Statistics Canada, National Accounts (Table 36-10-0104-01), 2023 estimates.

Planned vs. Actual Investment

- National accounts measure **actual investment**.
- AE model uses **planned investment**: what firms intend to invest.
- **Actual investment** = planned investment + **unplanned inventory changes**.
- **Why inventories matter**:
 - If sales are **lower than expected**: inventories build up unintentionally (firms reduce output).
 - If sales are **higher than expected**: inventories fall unintentionally (firms increase output).
- **Canadian example (2022)**: supply chain disruptions left Canadian auto dealers with record-low inventories — firms scrambled to increase output.

Planned vs. Actual Investment

- National accounts measure **actual investment**.
- AE model uses **planned investment**: what firms intend to invest.
- **Actual investment** = planned investment + **unplanned inventory changes**.
- **Why inventories matter**:
 - If sales are **lower than expected**: inventories build up unintentionally (firms reduce output).
 - If sales are **higher than expected**: inventories fall unintentionally (firms increase output).
- **Canadian example (2022)**: supply chain disruptions left Canadian auto dealers with record-low inventories — firms scrambled to increase output.

Planned vs. Actual Investment

- National accounts measure **actual investment**.
- AE model uses **planned investment**: what firms intend to invest.
- **Actual investment** = planned investment + **unplanned inventory changes**.
- **Why inventories matter:**
 - If sales are lower than expected: inventories build up unintentionally (firms reduce output).
 - If sales are higher than expected: inventories fall unintentionally (firms increase output).
- **Canadian example (2022)**: supply chain disruptions left Canadian auto dealers with record-low inventories — firms scrambled to increase output.

Planned vs. Actual Investment

- National accounts measure **actual investment**.
- AE model uses **planned investment**: what firms intend to invest.
- **Actual investment** = planned investment + **unplanned inventory changes**.
- **Why inventories matter**:
 - If sales are **lower than expected**: inventories build up unintentionally (firms reduce output).
 - If sales are **higher than expected**: inventories fall unintentionally (firms increase output).
- **Canadian example (2022)**: supply chain disruptions left Canadian auto dealers with record-low inventories — firms scrambled to increase output.

Planned vs. Actual Investment

- National accounts measure **actual investment**.
- AE model uses **planned investment**: what firms intend to invest.
- **Actual investment** = planned investment + **unplanned inventory changes**.
- **Why inventories matter:**
 - If sales are **lower than expected**: inventories build up unintentionally (firms reduce output).
 - If sales are **higher than expected**: inventories fall unintentionally (firms increase output).
- **Canadian example (2022)**: supply chain disruptions left Canadian auto dealers with record-low inventories — firms scrambled to increase output.

Planned vs. Actual Investment

- National accounts measure **actual investment**.
- AE model uses **planned investment**: what firms intend to invest.
- **Actual investment** = planned investment + **unplanned inventory changes**.
- **Why inventories matter**:
 - If sales are **lower than expected**: inventories build up unintentionally (firms reduce output).
 - If sales are **higher than expected**: inventories fall unintentionally (firms increase output).
- **Canadian example (2022)**: supply chain disruptions left Canadian auto dealers with record-low inventories — firms scrambled to increase output.

Planned vs. Actual Investment

- National accounts measure **actual investment**.
- AE model uses **planned investment**: what firms intend to invest.
- **Actual investment** = planned investment + **unplanned inventory changes**.
- **Why inventories matter**:
 - If sales are **lower than expected**: inventories build up unintentionally (firms reduce output).
 - If sales are **higher than expected**: inventories fall unintentionally (firms increase output).
- **Canadian example (2022)**: supply chain disruptions left Canadian auto dealers with record-low inventories — firms scrambled to increase output.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE > output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE < output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE > output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE < output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE > output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE < output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE > output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE < output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE > output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE < output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE > output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE < output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE $>$ output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE $<$ output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Macroeconomic Equilibrium in the AE Model

- **Macroeconomic equilibrium:**

$$\text{Planned AE} = \text{Real GDP}$$

- If planned AE $>$ output:
 - firms sell more than expected, inventories fall,
 - firms **increase** production and employment.
- If planned AE $<$ output:
 - firms sell less than expected, inventories rise,
 - firms **reduce** production and employment.
- **Inventories are the adjustment mechanism** that pushes the economy toward equilibrium.

Determinants of Consumption

Increase Consumption

- Higher disposable income
- Higher household wealth (house prices, stock market)
- Higher expected future income
- Lower interest rates
- Higher consumer confidence

Decrease Consumption

- Lower disposable income
 - Wealth destruction (market crash)
 - Pessimistic income expectations
 - Higher interest rates
 - Low consumer confidence
-
- **Canadian example (2022–2024):** Bank of Canada raised overnight rate from 0.25% to 5.0%. This reduced consumer spending, especially on mortgages, autos, and durables — Canadian consumer confidence fell sharply.

The Consumption Function and MPC

- The **consumption function**:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption (consumption when $Y_D = 0$),
 - Y_D = disposable income = $Y - T_{\text{net}}$,
 - MPC = marginal propensity to consume.
- **MPC:**
$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$
 - **Canadian estimate (Statistics Canada):** $\text{MPC} \approx 0.75\text{--}0.80$.
 - A \$1,000 tax rebate $\Rightarrow$ roughly \$750–\$800 in new spending.

The Consumption Function and MPC

- The **consumption function**:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption (consumption when $Y_D = 0$),
 - Y_D = disposable income = $Y - T_{\text{net}}$,
 - MPC = marginal propensity to consume.
- **MPC:**
$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$
 - **Canadian estimate (Statistics Canada):** $\text{MPC} \approx 0.75\text{--}0.80$.
 - A \$1,000 tax rebate $\Rightarrow$ roughly \$750–\$800 in new spending.

The Consumption Function and MPC

- The **consumption function**:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption (consumption when $Y_D = 0$),
 - Y_D = disposable income = $Y - T_{\text{net}}$,
 - MPC = marginal propensity to consume.
- **MPC:**
$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$
 - **Canadian estimate (Statistics Canada):** $\text{MPC} \approx 0.75\text{--}0.80$.
 - A \$1,000 tax rebate $\Rightarrow$ roughly \$750–\$800 in new spending.

The Consumption Function and MPC

- The **consumption function**:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption (consumption when $Y_D = 0$),
 - Y_D = disposable income = $Y - T_{\text{net}}$,
 - MPC = marginal propensity to consume.
- **MPC:**
$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$
 - **Canadian estimate (Statistics Canada):** $\text{MPC} \approx 0.75\text{--}0.80$.
 - A \$1,000 tax rebate $\Rightarrow$ roughly \$750–\$800 in new spending.

The Consumption Function and MPC

- The **consumption function**:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption (consumption when $Y_D = 0$),
 - Y_D = disposable income = $Y - T_{\text{net}}$,
 - MPC = marginal propensity to consume.
- **MPC:**
$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$
 - **Canadian estimate (Statistics Canada):** $\text{MPC} \approx 0.75\text{--}0.80$.
 - A \$1,000 tax rebate $\Rightarrow$ roughly \$750–\$800 in new spending.

The Consumption Function and MPC

- The **consumption function**:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption (consumption when $Y_D = 0$),
 - Y_D = disposable income = $Y - T_{\text{net}}$,
 - MPC = marginal propensity to consume.
- **MPC:**
- $$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$
- **Canadian estimate (Statistics Canada):** $\text{MPC} \approx 0.75\text{--}0.80$.
 - A \$1,000 tax rebate $\Rightarrow$ roughly \$750–\$800 in new spending.

The Consumption Function and MPC

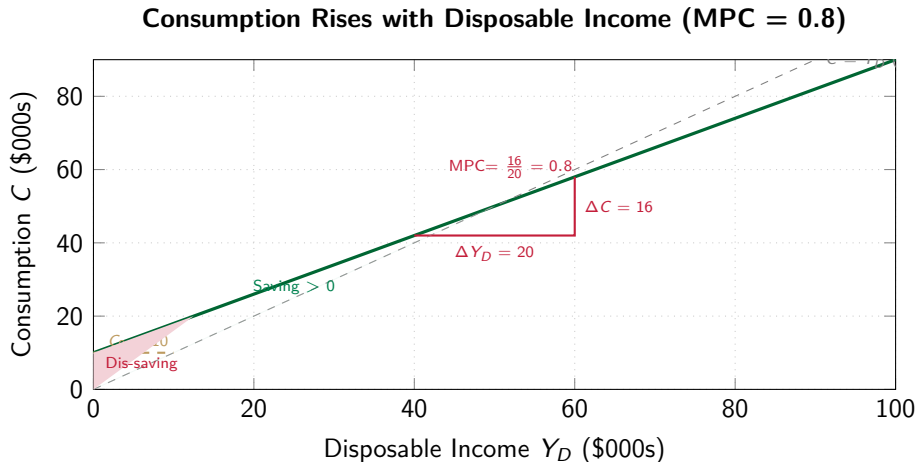
- The **consumption function**:

$$C = C_0 + \text{MPC} \times Y_D,$$

where:

- C_0 = autonomous consumption (consumption when $Y_D = 0$),
 - Y_D = disposable income = $Y - T_{\text{net}}$,
 - MPC = marginal propensity to consume.
- **MPC:**
$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}.$$
 - **Canadian estimate (Statistics Canada):** $\text{MPC} \approx 0.75\text{--}0.80$.
 - A \$1,000 tax rebate $\Rightarrow$ roughly **\$750–\$800** in new spending.

Figure 8.1 – The Consumption Function



MPC and MPS

- **Marginal propensity to consume (MPC):**

$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}$$

- Marginal propensity to save (MPS):

$$\text{MPS} = \frac{\Delta S}{\Delta Y_D} = 1 - \text{MPC}$$

- Every extra dollar of income is either spent or saved:

$$\text{MPC} + \text{MPS} = 1.$$

- **Example:** $\text{MPC} = 0.8$, so $\text{MPS} = 0.2$.
 - Of an extra \$1,000 income: \$800 spent, \$200 saved.

MPC and MPS

- **Marginal propensity to consume (MPC):**

$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}$$

- **Marginal propensity to save (MPS):**

$$\text{MPS} = \frac{\Delta S}{\Delta Y_D} = 1 - \text{MPC}$$

- Every extra dollar of income is either spent or saved:

$$\text{MPC} + \text{MPS} = 1.$$

- **Example:** $\text{MPC} = 0.8$, so $\text{MPS} = 0.2$.
 - Of an extra \$1,000 income: \$800 spent, \$200 saved.

MPC and MPS

- **Marginal propensity to consume (MPC):**

$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}$$

- **Marginal propensity to save (MPS):**

$$\text{MPS} = \frac{\Delta S}{\Delta Y_D} = 1 - \text{MPC}$$

- Every extra dollar of income is either spent or saved:

$$\text{MPC} + \text{MPS} = 1.$$

- **Example:** $\text{MPC} = 0.8$, so $\text{MPS} = 0.2$.
 - Of an extra \$1,000 income: \$800 spent, \$200 saved.

MPC and MPS

- **Marginal propensity to consume (MPC):**

$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}$$

- **Marginal propensity to save (MPS):**

$$\text{MPS} = \frac{\Delta S}{\Delta Y_D} = 1 - \text{MPC}$$

- Every extra dollar of income is either spent or saved:

$$\text{MPC} + \text{MPS} = 1.$$

- **Example:** $\text{MPC} = 0.8$, so $\text{MPS} = 0.2$.
 - Of an extra \$1,000 income: \$800 spent, \$200 saved.

MPC and MPS

- **Marginal propensity to consume (MPC):**

$$\text{MPC} = \frac{\Delta C}{\Delta Y_D}$$

- **Marginal propensity to save (MPS):**

$$\text{MPS} = \frac{\Delta S}{\Delta Y_D} = 1 - \text{MPC}$$

- Every extra dollar of income is either spent or saved:

$$\text{MPC} + \text{MPS} = 1.$$

- **Example:** $\text{MPC} = 0.8$, so $\text{MPS} = 0.2$.
 - Of an extra \$1,000 income: \$800 spent, \$200 saved.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Planned Investment

- **Expectations of future profits:**

- Firms invest when optimistic. Recessions reduce planned investment.
- **Canada 2020:** business investment fell $\approx 15\%$ due to pandemic uncertainty.

- **Real interest rate:**

- Higher rates $\Rightarrow$ higher cost of borrowing $\Rightarrow$ less investment.
- **Canada 2022–23:** BoC hikes pushed mortgage rates to 6–7%, residential construction fell sharply.

- **Taxes:**

- Higher corporate tax rates reduce after-tax returns on investment.
- Canada's federal corporate rate: 15% (base) + provincial rate.

- **Cash flow:**

- Many firms finance investment internally. Lower profits $\Rightarrow$ less cash $\Rightarrow$ less investment.

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- Determinants:
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - **CAD appreciates (CAD strengthens):** Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - **CAD depreciates (CAD weakens):** Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- **Determinants:**
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - **CAD appreciates** (CAD strengthens): Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - **CAD depreciates** (CAD weakens): Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- **Determinants:**
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - **CAD appreciates** (CAD strengthens): Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - **CAD depreciates** (CAD weakens): Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- Determinants:
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - CAD appreciates (CAD strengthens): Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - CAD depreciates (CAD weakens): Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- Determinants:
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - **CAD appreciates** (CAD strengthens): Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - **CAD depreciates** (CAD weakens): Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- Determinants:
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - **CAD appreciates** (CAD strengthens): Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - **CAD depreciates** (CAD weakens): Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- Determinants:
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - **CAD appreciates** (CAD strengthens): Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - **CAD depreciates** (CAD weakens): Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

Determinants of Net Exports (NX)

- **Net exports** = exports – imports.
- Determinants:
 - **Relative price levels:** if Canadian prices rise faster than foreign prices, exports fall and imports rise $\Rightarrow$ NX falls.
 - **Relative growth rates:** if Canada grows faster than trading partners, imports rise $\Rightarrow$ NX falls.
 - **Exchange rate (CAD/USD):**
 - **CAD appreciates** (CAD strengthens): Canadian exports more expensive for foreigners; imports cheaper $\Rightarrow$ NX falls.
 - **CAD depreciates** (CAD weakens): Canadian exports cheaper; imports more expensive $\Rightarrow$ NX rises.
- **Canada's key trading partner:** the U.S. takes $\approx 75\%$ of Canadian exports (Statistics Canada, 2024).

8.3 Learning Objective

Goal

Use a 45°-line diagram to illustrate macroeconomic equilibrium.

- Graph **real GDP** on the horizontal axis.
- Graph **planned aggregate expenditure** on the vertical axis.
- The **45°line** shows all points where $AE = Y$ (equilibrium).
- The AE line is **upward sloping** with slope = MPC (since higher $Y \Rightarrow$ higher $C \Rightarrow$ higher AE).

8.3 Learning Objective

Goal

Use a 45°-line diagram to illustrate macroeconomic equilibrium.

- Graph **real GDP** on the horizontal axis.
- Graph **planned aggregate expenditure** on the vertical axis.
- The **45°line** shows all points where $AE = Y$ (equilibrium).
- The AE line is **upward sloping** with slope = MPC (since higher $Y \Rightarrow$ higher $C \Rightarrow$ higher AE).

8.3 Learning Objective

Goal

Use a 45°-line diagram to illustrate macroeconomic equilibrium.

- Graph **real GDP** on the horizontal axis.
- Graph **planned aggregate expenditure** on the vertical axis.
- The **45°line** shows all points where $AE = Y$ (equilibrium).
- The AE line is **upward sloping** with slope = MPC (since higher $Y \Rightarrow$ higher $C \Rightarrow$ higher AE).

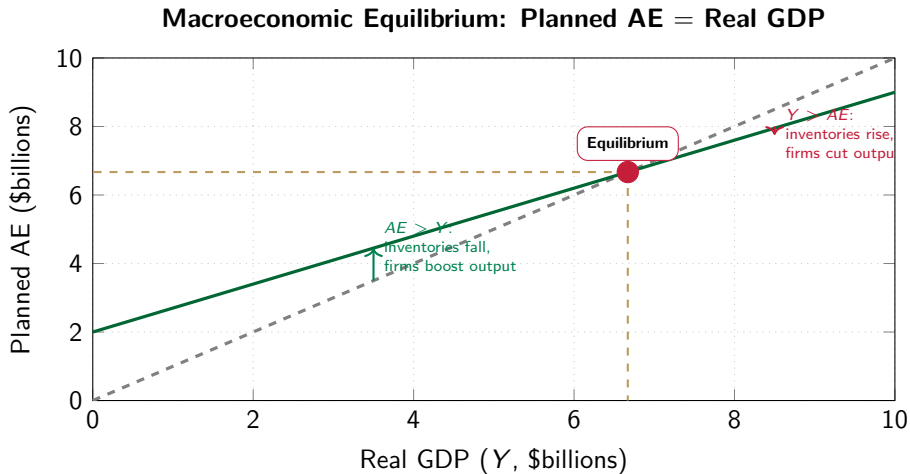
8.3 Learning Objective

Goal

Use a 45°-line diagram to illustrate macroeconomic equilibrium.

- Graph **real GDP** on the horizontal axis.
- Graph **planned aggregate expenditure** on the vertical axis.
- The **45°line** shows all points where $AE = Y$ (equilibrium).
- The AE line is **upward sloping** with slope = MPC (since higher $Y \Rightarrow$ higher $C \Rightarrow$ higher AE).

Figure 8.2 – The 45°-Line Diagram



Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

Understanding the 45°Line

- **On the 45°line:** Real GDP = Planned AE — equilibrium.
- **Above 45°line** ($AE > Y$):
 - Planned spending exceeds output.
 - Inventories are depleted.
 - Firms **increase** production $\Rightarrow$ GDP rises toward equilibrium.
 - **Example:** post-COVID reopening 2021 — consumers spent more than firms produced, pushing GDP higher.
- **Below 45°line** ($AE < Y$):
 - Planned spending is less than output.
 - Inventories accumulate unintentionally.
 - Firms **reduce** production $\Rightarrow$ GDP falls toward equilibrium.
 - **Example:** 2020 COVID lockdowns — businesses produced but few consumers could buy.

8.4 Learning Objective

Goal

Describe the multiplier effect and use the multiplier formula to calculate changes in equilibrium GDP.

- A small change in **autonomous expenditure** causes a **larger** change in equilibrium real GDP.
- This amplification is called the **multiplier effect**.
- Why? Because an initial increase in spending raises incomes, which raises consumption, which raises incomes further, and so on.

8.4 Learning Objective

Goal

Describe the multiplier effect and use the multiplier formula to calculate changes in equilibrium GDP.

- A small change in **autonomous expenditure** causes a **larger** change in equilibrium real GDP.
- This amplification is called the **multiplier effect**.
- Why? Because an initial increase in spending raises incomes, which raises consumption, which raises incomes further, and so on.

8.4 Learning Objective

Goal

Describe the multiplier effect and use the multiplier formula to calculate changes in equilibrium GDP.

- A small change in **autonomous expenditure** causes a **larger** change in equilibrium real GDP.
- This amplification is called the **multiplier effect**.
- Why? Because an initial increase in spending raises incomes, which raises consumption, which raises incomes further, and so on.

The Multiplier: Step-by-Step

- Suppose the federal government spends **\$10 billion** on new broadband infrastructure.
- **Round 1:** Telecom workers and contractors receive \$10B income.
 - They spend $MPC \times 10 = 0.8 \times 10 = \$8B$ on goods and services.
- **Round 2:** Retailers and service providers receive \$8B.
 - They spend $0.8 \times 8 = \$6.4B$.
- **Round 3:** $0.8 \times 6.4 = \$5.12B$, and so on...
- **Total effect:** $\Delta Y = 10 + 8 + 6.4 + 5.12 + \dots = \frac{1}{1-0.8} \times 10 = \textbf{\$50B}$.

The Multiplier: Step-by-Step

- Suppose the federal government spends **\$10 billion** on new broadband infrastructure.
- **Round 1:** Telecom workers and contractors receive \$10B income.
 - They spend $MPC \times 10 = 0.8 \times 10 = \$8B$ on goods and services.
- **Round 2:** Retailers and service providers receive \$8B.
 - They spend $0.8 \times 8 = \$6.4B$.
- **Round 3:** $0.8 \times 6.4 = \$5.12B$, and so on...
- **Total effect:** $\Delta Y = 10 + 8 + 6.4 + 5.12 + \dots = \frac{1}{1-0.8} \times 10 = \$50B$.

The Multiplier: Step-by-Step

- Suppose the federal government spends **\$10 billion** on new broadband infrastructure.
- **Round 1:** Telecom workers and contractors receive \$10B income.
 - They spend $MPC \times 10 = 0.8 \times 10 = \$8B$ on goods and services.
- **Round 2:** Retailers and service providers receive \$8B.
 - They spend $0.8 \times 8 = \$6.4B$.
- **Round 3:** $0.8 \times 6.4 = \$5.12B$, and so on...
- **Total effect:** $\Delta Y = 10 + 8 + 6.4 + 5.12 + \dots = \frac{1}{1-0.8} \times 10 = \textbf{\$50B}$.

The Multiplier: Step-by-Step

- Suppose the federal government spends **\$10 billion** on new broadband infrastructure.
- **Round 1:** Telecom workers and contractors receive \$10B income.
 - They spend $MPC \times 10 = 0.8 \times 10 = \$8B$ on goods and services.
- **Round 2:** Retailers and service providers receive \$8B.
 - They spend $0.8 \times 8 = \$6.4B$.
- **Round 3:** $0.8 \times 6.4 = \$5.12B$, and so on...
- **Total effect:** $\Delta Y = 10 + 8 + 6.4 + 5.12 + \dots = \frac{1}{1-0.8} \times 10 = \$50B$.

The Multiplier: Step-by-Step

- Suppose the federal government spends **\$10 billion** on new broadband infrastructure.
- **Round 1:** Telecom workers and contractors receive \$10B income.
 - They spend $MPC \times 10 = 0.8 \times 10 = \$8B$ on goods and services.
- **Round 2:** Retailers and service providers receive \$8B.
 - They spend $0.8 \times 8 = \$6.4B$.
- **Round 3:** $0.8 \times 6.4 = \$5.12B$, and so on...
- **Total effect:** $\Delta Y = 10 + 8 + 6.4 + 5.12 + \dots = \frac{1}{1-0.8} \times 10 = \$50B$.

The Multiplier: Step-by-Step

- Suppose the federal government spends **\$10 billion** on new broadband infrastructure.
- **Round 1:** Telecom workers and contractors receive \$10B income.
 - They spend $MPC \times 10 = 0.8 \times 10 = \$8B$ on goods and services.
- **Round 2:** Retailers and service providers receive \$8B.
 - They spend $0.8 \times 8 = \$6.4B$.
- **Round 3:** $0.8 \times 6.4 = \$5.12B$, and so on...
- **Total effect:** $\Delta Y = 10 + 8 + 6.4 + 5.12 + \dots = \frac{1}{1-0.8} \times 10 = \$50B$.

The Multiplier: Step-by-Step

- Suppose the federal government spends **\$10 billion** on new broadband infrastructure.
- **Round 1:** Telecom workers and contractors receive \$10B income.
 - They spend $MPC \times 10 = 0.8 \times 10 = \$8B$ on goods and services.
- **Round 2:** Retailers and service providers receive \$8B.
 - They spend $0.8 \times 8 = \$6.4B$.
- **Round 3:** $0.8 \times 6.4 = \$5.12B$, and so on...
- **Total effect:** $\Delta Y = 10 + 8 + 6.4 + 5.12 + \dots = \frac{1}{1-0.8} \times 10 = \mathbf{\$50B}$.

Figure 8.3 – The Multiplier Effect (MPC = 0.8)

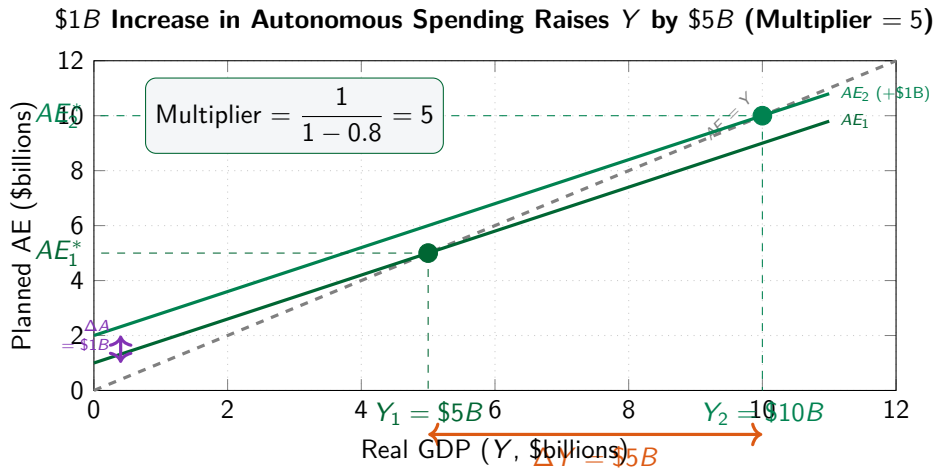
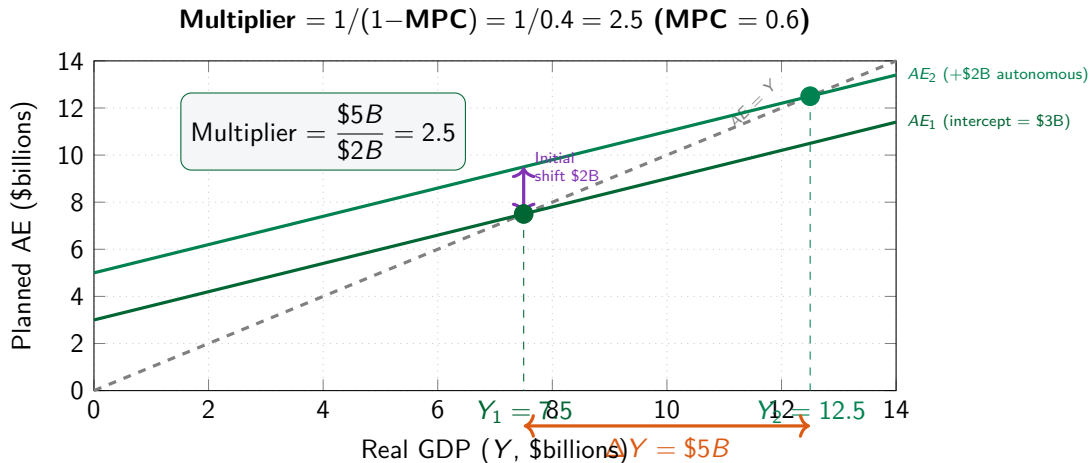


Figure 8.3b – Multiplier Effect (Cleaner Illustration)



Formula for the Multiplier

- The total change in equilibrium GDP from an initial change ΔA in autonomous expenditure:

$$\Delta Y = \Delta A \left[1 + \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots \right] = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- Multiplier:**

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

- Examples:**

- $\text{MPC} = 0.8$: Multiplier $= \frac{1}{0.2} = 5$.
- $\text{MPC} = 0.75$: Multiplier $= \frac{1}{0.25} = 4$.
- $\text{MPC} = 0.5$: Multiplier $= \frac{1}{0.5} = 2$.

- Real-world multiplier:** much smaller (1.0–1.5) due to taxes, imports, and price changes.

Formula for the Multiplier

- The total change in equilibrium GDP from an initial change ΔA in autonomous expenditure:

$$\Delta Y = \Delta A [1 + \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots] = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- **Multiplier:**

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

- **Examples:**

- $\text{MPC} = 0.8$: $\text{Multiplier} = \frac{1}{0.2} = 5.$
- $\text{MPC} = 0.75$: $\text{Multiplier} = \frac{1}{0.25} = 4.$
- $\text{MPC} = 0.5$: $\text{Multiplier} = \frac{1}{0.5} = 2.$

- **Real-world multiplier:** much smaller (1.0–1.5) due to taxes, imports, and price changes.

Formula for the Multiplier

- The total change in equilibrium GDP from an initial change ΔA in autonomous expenditure:

$$\Delta Y = \Delta A \left[1 + \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots \right] = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- **Multiplier:**

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

- **Examples:**

- $\text{MPC} = 0.8$: $\text{Multiplier} = \frac{1}{0.2} = 5$.
- $\text{MPC} = 0.75$: $\text{Multiplier} = \frac{1}{0.25} = 4$.
- $\text{MPC} = 0.5$: $\text{Multiplier} = \frac{1}{0.5} = 2$.
- **Real-world multiplier:** much smaller (1.0–1.5) due to taxes, imports, and price changes.

Formula for the Multiplier

- The total change in equilibrium GDP from an initial change ΔA in autonomous expenditure:

$$\Delta Y = \Delta A [1 + \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots] = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- **Multiplier:**

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

- **Examples:**

- $\text{MPC} = 0.8$: $\text{Multiplier} = \frac{1}{0.2} = 5$.
- $\text{MPC} = 0.75$: $\text{Multiplier} = \frac{1}{0.25} = 4$.
- $\text{MPC} = 0.5$: $\text{Multiplier} = \frac{1}{0.5} = 2$.
- **Real-world multiplier:** much smaller (1.0–1.5) due to taxes, imports, and price changes.

Formula for the Multiplier

- The total change in equilibrium GDP from an initial change ΔA in autonomous expenditure:

$$\Delta Y = \Delta A \left[1 + \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots \right] = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- **Multiplier:**

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

- **Examples:**

- $\text{MPC} = 0.8$: $\text{Multiplier} = \frac{1}{0.2} = 5$.
- $\text{MPC} = 0.75$: $\text{Multiplier} = \frac{1}{0.25} = 4$.
- $\text{MPC} = 0.5$: $\text{Multiplier} = \frac{1}{0.5} = 2$.
- **Real-world multiplier:** much smaller (1.0–1.5) due to taxes, imports, and price changes.

Formula for the Multiplier

- The total change in equilibrium GDP from an initial change ΔA in autonomous expenditure:

$$\Delta Y = \Delta A \left[1 + \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots \right] = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- **Multiplier:**

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

- **Examples:**

- $\text{MPC} = 0.8$: $\text{Multiplier} = \frac{1}{0.2} = 5$.
- $\text{MPC} = 0.75$: $\text{Multiplier} = \frac{1}{0.25} = 4$.
- $\text{MPC} = 0.5$: $\text{Multiplier} = \frac{1}{0.5} = 2$.
- **Real-world multiplier:** much smaller (1.0–1.5) due to taxes, imports, and price changes.

Formula for the Multiplier

- The total change in equilibrium GDP from an initial change ΔA in autonomous expenditure:

$$\Delta Y = \Delta A \left[1 + \text{MPC} + \text{MPC}^2 + \text{MPC}^3 + \dots \right] = \frac{1}{1 - \text{MPC}} \times \Delta A.$$

- **Multiplier:**

$$\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{\text{MPS}}$$

- **Examples:**

- $\text{MPC} = 0.8$: $\text{Multiplier} = \frac{1}{0.2} = 5$.
- $\text{MPC} = 0.75$: $\text{Multiplier} = \frac{1}{0.25} = 4$.
- $\text{MPC} = 0.5$: $\text{Multiplier} = \frac{1}{0.5} = 2$.
- **Real-world multiplier:** much smaller (1.0–1.5) due to taxes, imports, and price changes.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to 28% during lockdowns.
 - GDP fell sharply (demand shock).
 - The “forced saving” was then partially spent during the 2021–22 reopening boom.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to 28% during lockdowns.
 - GDP fell sharply (demand shock).
 - The “forced saving” was then partially spent during the 2021–22 reopening boom.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to 28% during lockdowns.
 - GDP fell sharply (demand shock).
 - The "forced saving" was then partially spent during the 2021–22 reopening boom.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to 28% during lockdowns.
 - GDP fell sharply (demand shock).
 - The “forced saving” was then partially spent during the 2021–22 reopening boom.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to 28% during lockdowns.
 - GDP fell sharply (demand shock).
 - The “forced saving” was then partially spent during the 2021–22 reopening boom.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to 28% during lockdowns.
 - GDP fell sharply (demand shock).
 - The “forced saving” was then partially spent during the 2021–22 reopening boom.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to 28% during lockdowns.
 - GDP fell sharply (demand shock).
 - The “forced saving” was then partially spent during the 2021–22 reopening boom.

The Paradox of Thrift

- **Long run:** higher saving $\Rightarrow$ more investment $\Rightarrow$ higher growth.
- **Short run (AE model):** if all households simultaneously save more (consume less):
 - Consumption falls, AE falls, equilibrium GDP falls.
 - Could trigger a recession — even though each individual was being prudent.
- This is the **paradox of thrift**: what is good for individuals can be bad for the economy in the short run.
- **Canadian example (2020):** household saving rate surged to 28% during lockdowns.
 - GDP fell sharply (demand shock).
 - The “forced saving” was then partially spent during the 2021–22 reopening boom.

8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far: price level fixed.
- Now: what happens to AE when the **price level changes**?
- Three channels link a higher price level to lower AE:
 1. **Wealth effect** $\Rightarrow$ lower C .
 2. **Interest-rate effect** $\Rightarrow$ lower I .
 3. **International-trade effect** $\Rightarrow$ lower NX .
- A higher price level $\Rightarrow$ lower equilibrium GDP $\Rightarrow$ downward-sloping **aggregate demand curve**.

8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far: price level fixed.
- Now: what happens to AE when the **price level changes**?
- Three channels link a higher price level to lower AE:
 1. Wealth effect $\Rightarrow$ lower C .
 2. Interest-rate effect $\Rightarrow$ lower I .
 3. International-trade effect $\Rightarrow$ lower NX .
- A higher price level $\Rightarrow$ lower equilibrium GDP $\Rightarrow$ downward-sloping **aggregate demand curve**.

8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far: price level fixed.
- Now: what happens to AE when the **price level changes**?
- Three channels link a higher price level to lower AE:
 1. **Wealth effect** $\Rightarrow$ lower C .
 2. **Interest-rate effect** $\Rightarrow$ lower I .
 3. **International-trade effect** $\Rightarrow$ lower NX .
- A higher price level $\Rightarrow$ lower equilibrium GDP $\Rightarrow$ downward-sloping **aggregate demand curve**.

8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far: price level fixed.
- Now: what happens to AE when the **price level changes**?
- Three channels link a higher price level to lower AE:
 1. **Wealth effect** $\Rightarrow$ lower C .
 2. **Interest-rate effect** $\Rightarrow$ lower I .
 3. **International-trade effect** $\Rightarrow$ lower NX .
- A higher price level $\Rightarrow$ lower equilibrium GDP $\Rightarrow$ downward-sloping **aggregate demand curve**.

8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far: price level fixed.
- Now: what happens to AE when the **price level changes**?
- Three channels link a higher price level to lower AE:
 1. **Wealth effect** $\Rightarrow$ lower C .
 2. **Interest-rate effect** $\Rightarrow$ lower I .
 3. **International-trade effect** $\Rightarrow$ lower NX .
- A higher price level $\Rightarrow$ lower equilibrium GDP $\Rightarrow$ downward-sloping **aggregate demand curve**.

8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far: price level fixed.
- Now: what happens to AE when the **price level changes**?
- Three channels link a higher price level to lower AE:
 1. **Wealth effect** $\Rightarrow$ lower C .
 2. **Interest-rate effect** $\Rightarrow$ lower I .
 3. **International-trade effect** $\Rightarrow$ lower NX .
- A higher price level $\Rightarrow$ lower equilibrium GDP $\Rightarrow$ downward-sloping **aggregate demand curve**.

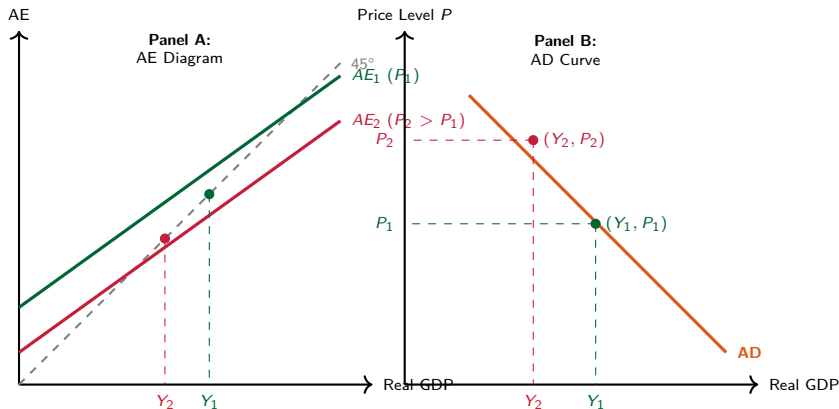
8.5 Learning Objective

Goal

Understand the relationship between the aggregate demand curve and aggregate expenditure.

- So far: price level fixed.
- Now: what happens to AE when the **price level changes**?
- Three channels link a higher price level to lower AE:
 1. **Wealth effect** $\Rightarrow$ lower C .
 2. **Interest-rate effect** $\Rightarrow$ lower I .
 3. **International-trade effect** $\Rightarrow$ lower NX .
- A higher price level $\Rightarrow$ lower equilibrium GDP $\Rightarrow$ downward-sloping **aggregate demand curve**.

Figure 8.4 – From AE to the Aggregate Demand Curve



Higher price level $\Rightarrow$ AE shifts down $\Rightarrow$ lower equilibrium $Y \Rightarrow$ downward-sloping AD.

Appendix – Solving the AE Model Algebraically

- Suppose the Canadian economy is described by:

$$\begin{aligned}C &= 400 + 0.8(Y - T), \\I &= 300, \quad G = 500, \quad T = 250, \quad NX = -50.\end{aligned}$$

- Equilibrium condition: $Y = C + I + G + NX$.
- Substituting:

$$Y = 400 + 0.8(Y - 250) + 300 + 500 - 50.$$

$$Y = 400 + 0.8Y - 200 + 750 \Rightarrow 0.2Y = 950 \Rightarrow Y = \$4,750 \text{ billion}.$$

- General multiplier: $\frac{1}{1-MPC} = \frac{1}{1-0.8} = 5$.

Appendix – Solving the AE Model Algebraically

- Suppose the Canadian economy is described by:

$$\begin{aligned}C &= 400 + 0.8(Y - T), \\I &= 300, \quad G = 500, \quad T = 250, \quad NX = -50.\end{aligned}$$

- Equilibrium condition: $Y = C + I + G + NX$.

- Substituting:

$$Y = 400 + 0.8(Y - 250) + 300 + 500 - 50.$$

$$Y = 400 + 0.8Y - 200 + 750 \Rightarrow 0.2Y = 950 \Rightarrow Y = \$4,750 \text{ billion}.$$

- General multiplier: $\frac{1}{1-MPC} = \frac{1}{1-0.8} = 5$.

Appendix – Solving the AE Model Algebraically

- Suppose the Canadian economy is described by:

$$\begin{aligned}C &= 400 + 0.8(Y - T), \\I &= 300, \quad G = 500, \quad T = 250, \quad NX = -50.\end{aligned}$$

- Equilibrium condition: $Y = C + I + G + NX$.
- Substituting:

$$Y = 400 + 0.8(Y - 250) + 300 + 500 - 50.$$

$$Y = 400 + 0.8Y - 200 + 750 \Rightarrow 0.2Y = 950 \Rightarrow \boxed{Y = \$4,750 \text{ billion}}.$$

- General multiplier: $\frac{1}{1-MPC} = \frac{1}{1-0.8} = 5$.

Appendix – Solving the AE Model Algebraically

- Suppose the Canadian economy is described by:

$$\begin{aligned}C &= 400 + 0.8(Y - T), \\I &= 300, \quad G = 500, \quad T = 250, \quad NX = -50.\end{aligned}$$

- Equilibrium condition: $Y = C + I + G + NX$.
- Substituting:

$$Y = 400 + 0.8(Y - 250) + 300 + 500 - 50.$$

$$Y = 400 + 0.8Y - 200 + 750 \Rightarrow 0.2Y = 950 \Rightarrow Y = \$4,750 \text{ billion}.$$

- General multiplier: $\frac{1}{1-MPC} = \frac{1}{1-0.8} = 5$.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Summary

- The **AE model** determines short-run equilibrium when the price level is fixed.
- Equilibrium: Planned AE = Y; inventories adjust to reach this point.
- **Consumption** driven by disposable income ($MPC \approx 0.75\text{--}0.80$ in Canada), wealth, expectations, and interest rates.
- **Planned investment** driven by profit expectations, interest rates, taxes, and cash flow.
- **Net exports** driven by relative prices, relative growth rates, and the exchange rate.
- The **multiplier** = $1/(1 - MPC)$ amplifies changes in autonomous spending:
 - Works in both directions — stimulus boosts GDP, contractionary shock reduces it.
 - Real-world multipliers are smaller due to taxes, imports, and price changes.
- The **paradox of thrift**: individually rational saving can reduce GDP in the short run.
- As the price level changes, AE shifts, tracing out the downward-sloping **aggregate demand curve**.

Thank you!

Questions?